

Deep Multi-Objective Reinforcement Learning for Utility-Based Infrastructural Maintenance Optimization

Jesse van Remmerden, Maurice Kenter, Diederik M. Roijers, Charalampos Andriotis, Yingqian Zhang & Zaharah Bukhsh
contact: j.v.remmerden@tue.nl

Problem Setting:

- We must plan maintenance, at the component level, for infrastructural assets, such as bridges.
- The problem is **multi-objective**, we want to minimize **cost** and **probability of failure**?
- How do we incorporate existing methodologies for weighing these objectives?

Multi-Objective Reinforcement Learning:

Instead of a scalar reward value, in MORL, we have a reward vector for each timestep $\vec{R}_t \in \mathbb{R}^d$, where d is the number of objectives.

How do we weigh objectives against each other? The episodic return $\vec{R} = \sum_{t=0}^H \gamma^t \vec{r}_t$ can be weighed using an utility function $u : \mathbb{R}^d \rightarrow \mathbb{R}$. Whereby the policy is updated through:

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[u \left(\sum_{k=t}^H \gamma^k \vec{r}_k \right) \mid \pi, s_t = s \right]. \quad (1)$$

Proposed Method: MO-DCMAC

We introduce Multi-Objective Deep Centralized Multi-Agent Actor-Critic (MO-DCMAC). Based on previous work DMCAC [1] and MOCAC [2]

The actor network uses a multi-head network architecture to output an action for each component. The critic learns a discrete multivariate probability distribution of the returns of each combination of the objectives. We use this to calculate a preference score:

$$u_t = \sum_{i \in \{1, \dots, \mathcal{N}\}^d} u(\vec{R}_t^- + \vec{z}_i) Z_{\psi}(\vec{z}_i \mid s_t), \quad (2)$$

where $\vec{R}_t^- = \sum_{k=0}^t \gamma^k \vec{r}_k$ is the accrued reward and $\vec{z}$ is the support vector representing all outcomes.

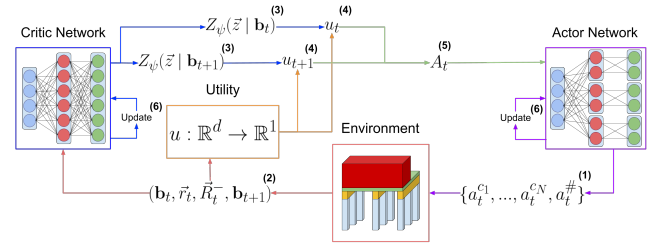


Figure 1: The figure illustrates the training sequence for MO-DCMAC.

Results:

Policy	Utility Score	Cost	Probability of Collapse
CBI-CBA	5.57 ± 3.54	1.49 ± 0.23	0.01 ± 0.02
YBI-CBA	9.57 ± 9.54	1.48 ± 0.2	0.04 ± 0.08
MO-DCMAC	5.07 ± 3.16	1.03 ± 0.16	0.02 ± 0.02

Table 1: Results with 13 components

Policy	Utility Score	Cost	Probability of Collapse
CBI-CBA	10.81 ± 7.18	1.45 ± 0.17	0.05 ± 0.05
YBI-CBA	16.04 ± 13.19	1.54 ± 0.14	0.08 ± 0.11
MO-DCMAC	7.29 ± 4.74	1.44 ± 0.28	0.02 ± 0.03

Table 2: Results with 26 components

References

- [1] Andriotis CP, Papakonstantinou KG (2019). Managing engineering systems with large state and action spaces through deep reinforcement learning. *Reliability Engineering and System Safety* **191**
- [2] Reymond M, Hayes CF, Steckelmacher D, et al (2023). Actor-critic multi-objective reinforcement learning for non-linear utility functions. *Autonomous Agents and Multi-Agent Systems* **37**.